

Activities of Information Security Research in Taiwan: TWISC and iCAST

Prof. Tzong-Chen Wu (吳宗成)

National Taiwan University of
Science & Technology (NTUST)

Agenda

- Introduction to TWISC
 - Organization
 - Research topics
- Introduction to iCAST
 - Organization
 - Research topics
- Activities hosted by TWISC/iCAST
 - Regular events
 - Upcoming events
- PiSA: Pervasive iCAST Security Angel
- Conclusions

About Dr. Tzong-Chen Wu

- Received BS from Department of Information Engineering, National Taiwan University, Taipei, Taiwan in June 1983, and MS from Institute of Applied Mathematics, National Chung Hsin University, Taichung, Taiwan, in June 1989
- Received PhD from Department of Computer Science & Information Engineering, National Chiao Tung University, Hsinchu, Taiwan, in June 1992
- Research interests: Cryptography, Information Security
- Current positions
 - Professor of Department of Information Management, NTUST, Taiwan (since Feb.1997)
 - Dean of School of Management, NTUST, Taiwan (since Aug. 2007)
 - President of Chinese Cryptology and Information Security Association (CCISA), Taiwan (since Aug. 2006)
 - CEO of Taiwan Information Security Center (TWISC), and Director of TWISC@NTUST (since April 2005)
 - PI of iCAST from Academic groups (Led by Dr.Der-Tsai Lee) (since June 2006)

Introduction to TWISC

Introduction to TWISC

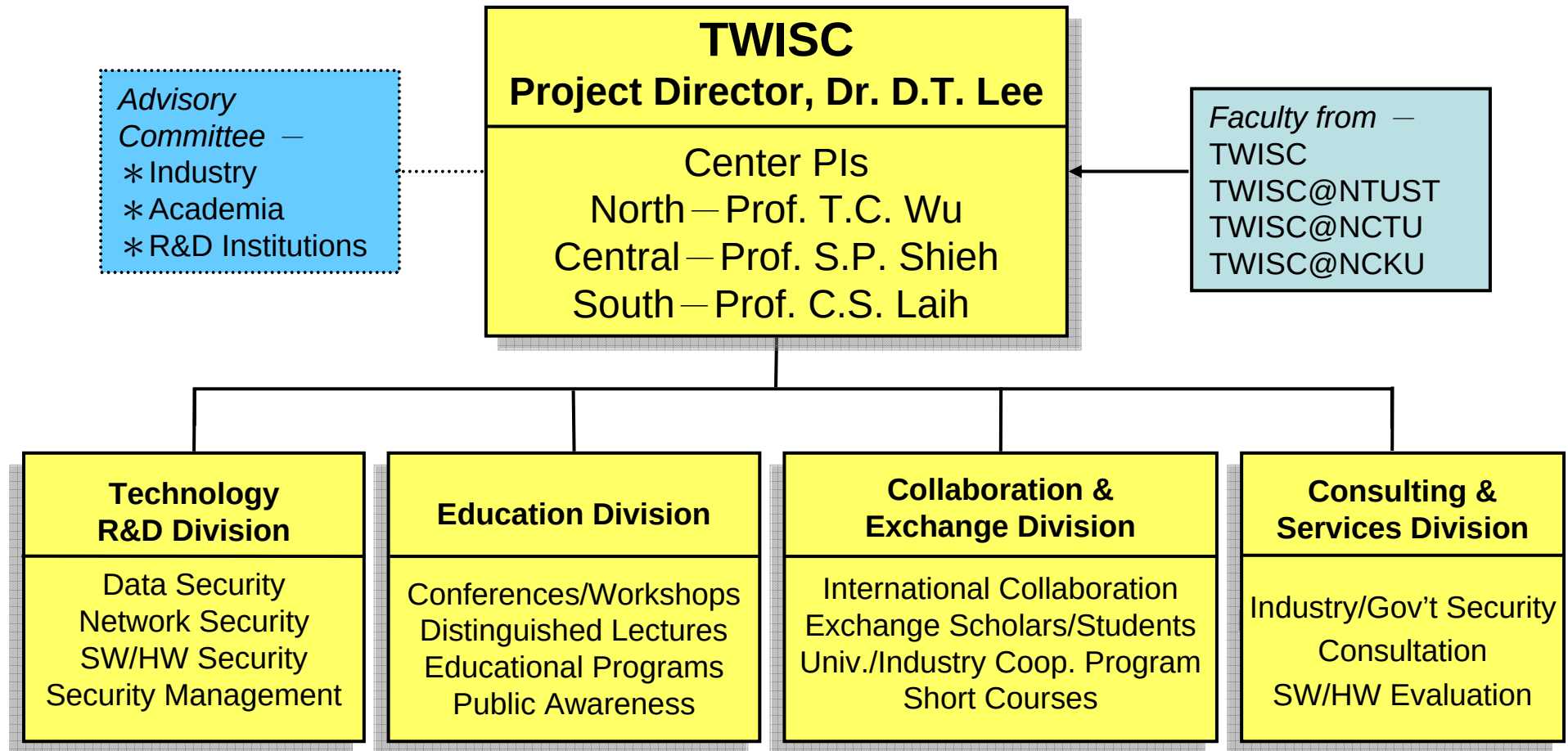
<http://www.twisc.org/>

- **TaiWan Information Security Center** was officially founded on April 1st, 2005, funded by National Science Council (NSC) of Taiwan (four-year project for academic group and 1 million USD per year)
- Missions of TWISC
 - Advance the research and development of technologies in information security and related areas
 - Provide education and training, help build human resource capacity, and promote public awareness in information security
 - Collaborate with private sectors to enhance local (Taiwan) information security industry in security management and applications software development
 - Seek international collaborations to build a ubiquitous secure community

Physical Location of TWISC

- Headquarter
 - Center for Information Technology Innovation, Academia Sinica (under construction)
 - Led by Dr. Der-Tsai Lee, Academician
- Three physical centers
 - Northern Taiwan: TWISC@NTUST
 - National Taiwan University of Science & Technology, since Nov. 2005
 - Central Taiwan: TWISC@NCTU
 - National Chiao Tung University, since Oct. 2006
 - Southern Taiwan: TWISC@NCKU
 - National Cheng Kung University, since Oct. 2006
 - Each center has more than 20 professors and 40 MS/PhD students joined

Functional Organization of TWISC



Research Topics

Coordinated by TWISC@NTUST

- Major topics
 - Cryptography
 - Authentication/digital signature, cryptographic protocols
 - Cryptographic techniques for low-resource/mobile devices
 - Multivariate public key cryptography
 - Software/Hardware security
 - Formal verification
 - Web-based application security assessment
 - Power analysis/hardware security assessment
 - Privacy protection and security management
 - Healthcare or ubiquitous applications
- Related topics
 - Smart card & RFID security
 - Network security

Research Topics

Coordinated by TWISC@NCTU

- Major topics
 - Wireless network security
 - IPv6 security
 - Cryptographic protocols
 - Sensor network security
 - Remote security scanner
- Related topics
 - Cryptography
 - Intrusion detection system
 - Information hiding & Watermarking
 - GSN-CERT (Government Service Network – Computer Emergency Response Team)

Research Topics

Coordinated by TWISC@NCKU

- Major topics
 - Intrusion detection system
 - IPv6-Enabled IDS
 - IPS for SIP VoIP System
 - DNS Server Protection
 - Testbed, Honeypot
 - More than 290 tools, and more than 20 testing scenarios
 - Integrate with EuroCom Honeypot network security statistics database
- Related topics
 - Cryptography
 - Digital forensics
 - Information hiding & Watermarking

Hosted Workshops and Conferences - SADFE 2005

- **SADFE 2005 (First International Workshop on Systematic Approaches to Digital Forensic Engineering)**, Nov. 7-10, 2005, Taipei, Taiwan
<http://conf.ncku.edu.tw/sadfe/>
- **Invited Speakers**
 - Prof. George Mohay: Queensland University of Technology
 - Prof. Steve Schroeder: Seattle University, a pioneer in cyber crime law enforcement
 - Dr. Ming-Yuh Huang: Boeing Associate Technical Fellow



Hosted Workshops and Conferences - AsiaCCS'06

- **ACM Symposium on Information, Computer and Communications Security (AsiaCCS'06)**, March 21-23, 2006, Taipei, Taiwan

<http://www.iis.sinica.edu.tw/asiaccs06/>

Distinguished Lecture Series

- Date: 2005/12/16
- Prof. **Hideki IMAI**
Information and Systems Department, Institute of Industrial Science
University of Tokyo
- Topic: Trends and Challenges for Securer Cryptography



Distinguished Lecture Series

Date: 2006/3/20

Prof. **Doug Tygar**, UC Berkeley

Prof. **Tzi-cker Chiueh**, State Uni. of New York at Stony Brook

Topics:

- (1) Open problems and promising approaches in computer security (Prof. Tygar)
- (2) Software Security and Solutions : A Stony Brook Perspective (Prof. Chiueh)



Distinguished Lecture Series

- Date: 2006/05/01
- Principle **Shigeo TSUJII**
Institute of Information Security, Japan
- Topic: Proposal for Piece in Hand (Soldiers in Hand) Matrix - General Method for Enhancing Security of Multivariate Public Key Cryptosystems

- Date: 2006/05/01
- Prof. **Yukio ITAKURA**
Institute of Information Security, Japan
- Topic: Recent Japanese Trend of Biometric Authentication Application System and Future ISEC Research Project



Distinguished Lecture Series

- Date: 2006/06/01
- Prof. **Kwangjo Kim**, ICU, Korea
- Topic: Security Challenges in Ubiquitous Society



Distinguished Lecture Series

- Date: 2006/07/06
- Prof. **Adrian Perrig**, CMU, USA
- Topic: Security in Sensor Network: Industry Trends, Present and Future Research Direction



Distinguished Lecture Series

- Date: 2006/08/04
- Prof. **Felix Wu**, UC Davis, USA
- Topic: Security and Robustness based on Nothing: A New Paradigm for Future Internet Design



Distinguished Lecture Series

- Date: 2006/11/24
- Prof. **Rei Safavi-Naini**
University of Wollongong, Australia
- Topic: Post-Quantum Cryptography



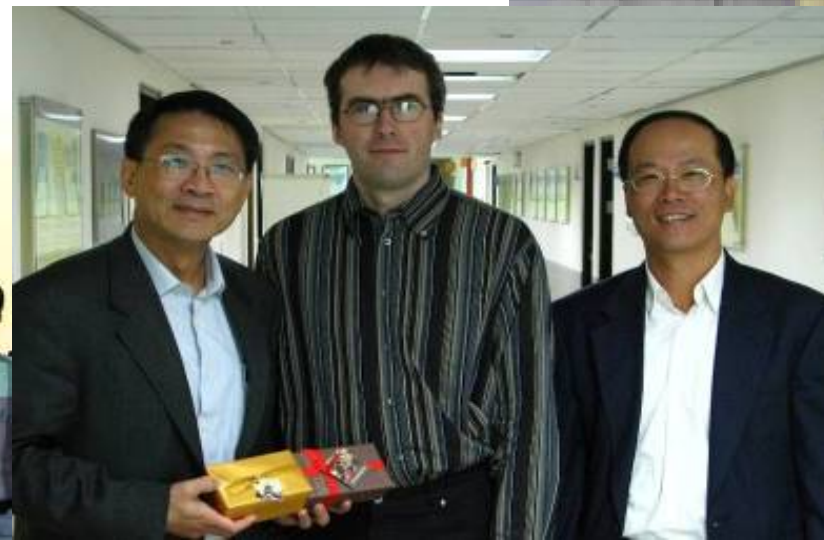
Distinguished Lecture Series

- Date: 2007/01/03
- Prof. **Horace Yuen**
Northwestern University, USA
- Topic: Quantum and Physical Cryptography



Distinguished Lecture Series

- Date: 2007/4/02
- Prof. **Vincent Rijmen**
Graz University of Technology
Institute for Applied Information Processing
and Communications (IAIK) - Krypto Group
- Topic: 10 years of Rijndael and Beyond



Recent Invited Speakers

- Dr. **Zhi-Jun Zhan**, CMU CyLab Japan
Privacy in digital age and privacy-preserving collaborative data mining
(2007/3/14)
- Mr. **Philip Ku**, TUV NORD, Taiwan Branch
Introduction to ISO/IEC 15408 – Evaluation and Applications (2007/4/23)
- Prof. **Kouichi SAKURAI**, Kyushu University
Cryptographic Infrastructures based on a Unique Device with Tamper-Resistant
(2007/6/13)

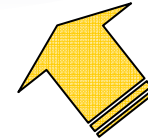
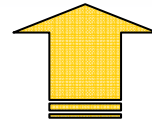
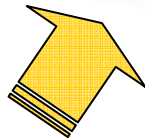


Workshop on Advanced Information Security Technology

Date: **2006/01/20-21** at NTUST

Topic: **Multivariate PKC**

Multivariate PKC



Prof. Jintai Ding, University
of Cincinnati, USA



Prof. Christopher Wolf, Ecole
Normale Supérieure, France



Prof. Bo-Yin Yang,
Academia Sinica,
Taiwan

Information Security Summer School (ISSS) Aug. 10~24, 2006

- **54-hour Lectures provided by:**
 - Prof. Tzong-Chen Wu (NTUST), Prof. Chi-Sung Lai (NCKU)
 - Prof. Bo-Yin Yang (Academia Sinica)
 - Prof. Dan Bernstein (U. Illinois Chicago)
 - Prof. Tanja Lange (Tech. U. Denmark)
 - Prof. Claus Diem (U. Leipzig Germany)
 - Dr. Rui Zhang (RCIS Japan)
 - Dr. Takayuki Miyadera (RCIS Japan)
 - Dr. Yutaka Oiwa (RCIS Japan)
 - Prof. Jason Crampton (ISG UK)
 - Prof. Keith Martin (ISG UK)
 - Prof. Chris Mitchell (ISG UK)
 - Prof. Keith Mayes (ISG UK)
- **Venue:** NTUST
- **For more information** see <http://www.twisc.org>

Information Security Summer School (ISSS) Aug. 6~16, 2007

- **54-hour Lectures provided by:**
 - Prof. Chris Mitchell (ISG)
 - Prof. Keith Mayes (ISG)
 - Dr. Marc Joye (Thomson)
 - Prof. Robert Deng (SMU)
 - Prof. Kwangjo Kim (ICU)
 - Prof. John Choi (Sangmyung)
 - Dr. Chan Yeob Yeun (LG)
 - Dr. Jonathan Millen (Mitre)
 - Prof. Naoshi Sato (IISec)
 - Prof. Tzong-Chen Wu (NTUST)
- **Venue: NTUST**
- **For more information see <http://www.twisc.org>**

The 11th Information Security Conference (ISC 2008)

- Venue: The Grand Hotel, Taipei, Taiwan
- Date: September, 2008 (TBD)
- Hosted by CCISA (Chinese Cryptology and Information Security Association), TWISC, Academia Sinica, NTUST, etc.
- Organizing Committee Members
 - Dr. Der-Tsai Lee (Academia Sinica)
 - Prof. Tzong-Chen Wu (NTUST)
 - Prof. Chin-Luang Lei (NTU)
 - Prof. Vincent Rijmen (Graz University of Technology, Austria)

Introduction to iCAST

Introduction to iCAST

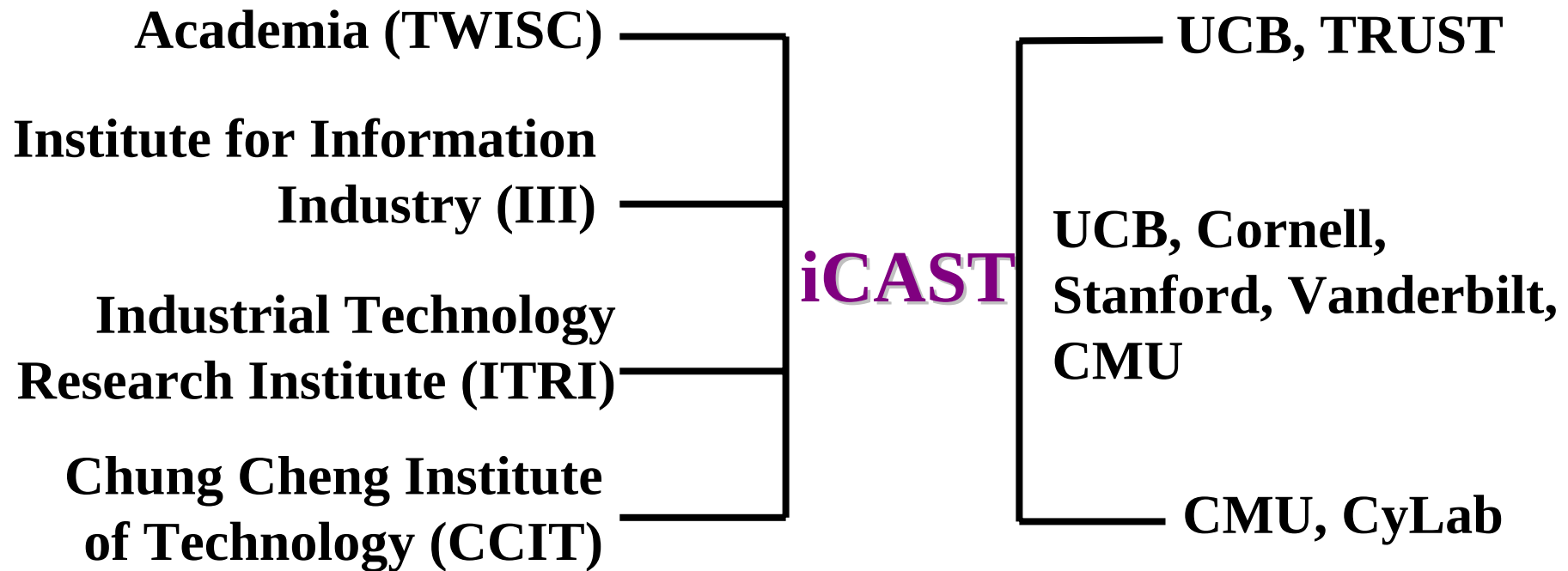
<http://www.icast.org.tw/>

- **International Collaboration for Advancing Security Technology (iCAST)** was officially founded on May 1st, 2006, funded by National Science Council (NSC) of Taiwan (three-year project for academic group coordinated by TWISC, and 1 million USD per year)
- A mission-oriented research team consisting of professors and researchers from:
 - Taiwan side: TWISC, Institute for Information Industry (III), Industrial Technology Research Institute (ITRI) and Chung Cheng Institute of Technology (CCIT)
 - US side: Carnegie Mellon University (CMU) CyLab and University of California, Berkeley (UCB) TRUST

Missions for iCAST

- Cultivate the seeds of information security and do academic works and practical implementations
- Dispatch professors and PhD students to CMU and UCB (about 15 person each year) for the short-term and long-term study to enlarge global vision in researches
- Objectives
 - To gain core technologies and develop practical prototypes jointly worked with CMU and UCB
 - To do technology transfer to the industries in Taiwan to promote their self-capability
 - To spread the knowledge of information security education

Functional Organization of iCAST



Academic Groups and Research Topics in iCAST

Subproject Title	Member	Collaborator
Security Technologies and Applications in Wireless Sensor Networks	5 Professors 11 P/M students	UC Berkeley TRUST
Security & Privacy Protection Technologies for RFID Applications	6 Professors 9 P/M students	UC Berkeley TRUST
Advanced Remote Authentication Technologies	6 Professors 6 P/M students	CMU CyLab
Investigation on Intrusion Detection Techniques	5 Professors 9 P/M students	CMU CyLab
Program/Software Security Evaluation Systems	3 Professors 8 P/M students	CMU & UCB
CMU-Realization of High Speed IDS Expert System UCB-Privacy-preserving Information Management	3 Professors 11 P/M students	CMU & UCB

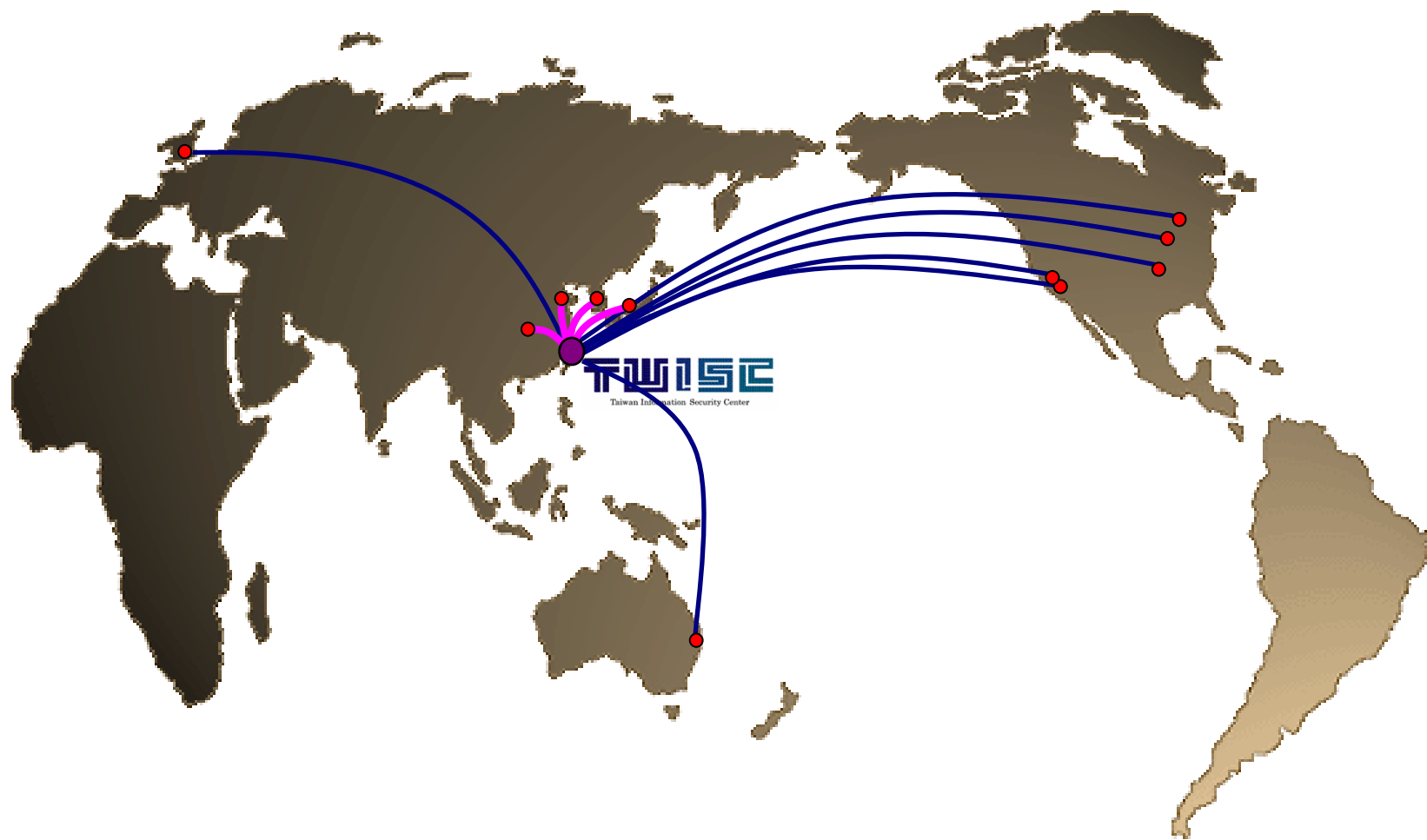
iCAST/CMU/TRUST Joint Conferences

- January 8-10, 2007, Taipei, Taiwan
- June 4-5, 2007, Taipei, Taiwan
- <http://www.icast.org.tw/events/conferences-and-workshops>

Developed Prototypes

- RISE (RFID + ITS + Sensor Network+ Embedded Systems)
 - A blood bag management system providing off-site tracking and on-site management (for hospital)
- Asset Monitoring and Protection
 - Record the flowing objects and prevent from stealing assets
- PiSA (Pervasive iCAST Security Angel)
 - Use PDA or cellular phone to authenticate the trustworthy website via visual random art
- IDEAS (Intrusion Detection and Event Analysis System)
 - Based on machine learning and AI methods to analyze real-time network attacks
- Security Monitor
 - Implemented on a Linux platform with a loadable kernel module to compute and update the probabilistic belief (inference engine) of the attack
- Personal Privacy Guard (OceanStore)
 - Provide personalized privacy preference rules generation, reasoning, and refinement for personal information protection

International Collaboration in TWISC

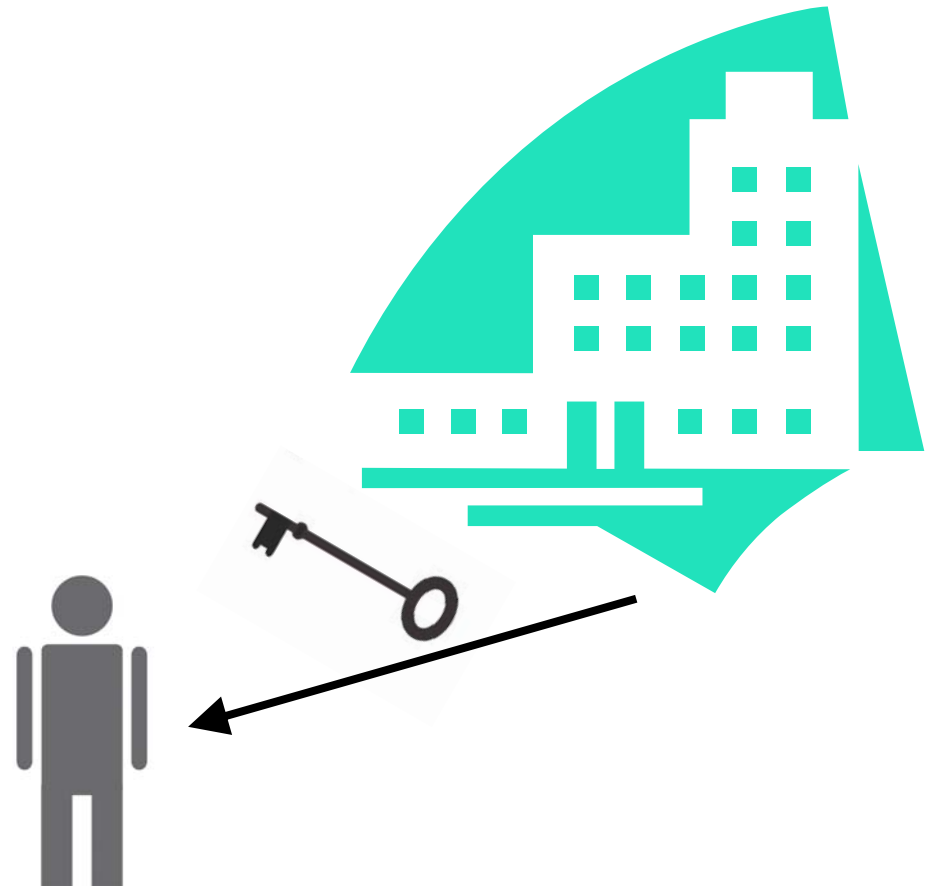


PiSA

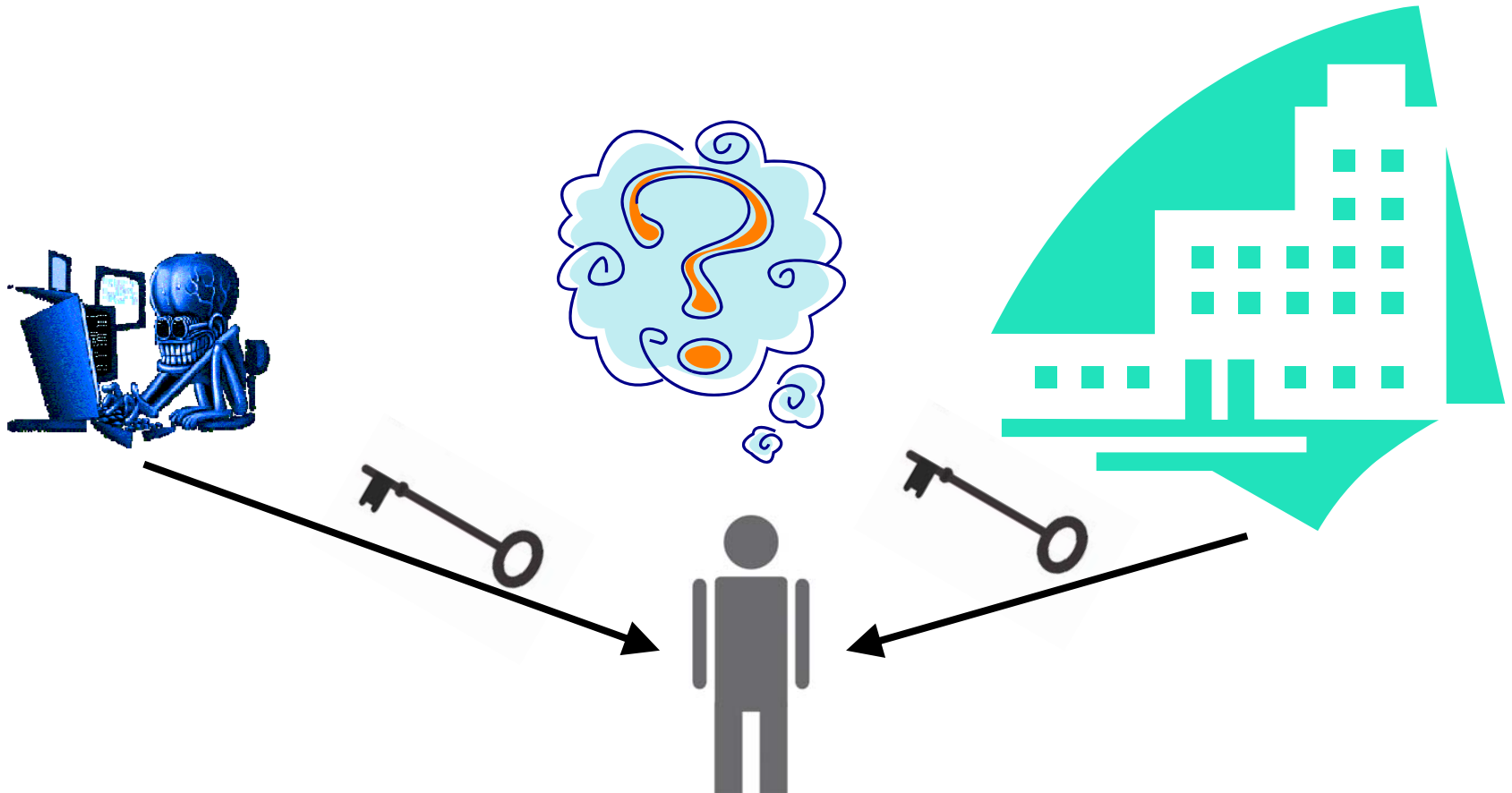
Pervasive iCAST Security Angel

Scenario for PiSA's Motivation

- I am arriving at JWIS2007 conference
- My mobile phone/device receives a public key and its certificate from JWIS2007, and requests to authenticate the wireless access to JWIS2007



Can I Trust This Key?



Functional Specification of PiSA

- Establish the framework to create trusted information with others we meet or communicate with, so PiSA should:
 - Verify integrity of environmental or infrastructure information
 - Use trusted information to build authenticated communication channel
 - Exchange a trusted crypto key with strangers (at least, untrusted party)
- Enable a user to discover trustworthy and authentic information about the environment
- Provide seamless integration with communication application, such that discovered trustworthy information is automatically employed by communication software, e.g., email, voice, IM

A Good Solution to Application

- Allow a people holding mobile device to authenticate the public key sending from an infrastructure (e.g., school, airport, coffee shop, company, etc.), and achieves the following requirements:
 - Secure
 - Easy to use
 - Unobtrusive
 - Fast
 - Scalable
 - Minimize specialized hardware
 - Evaluated by cost of hardware
 - Has low setup and maintenance cost
 - Easy to update key

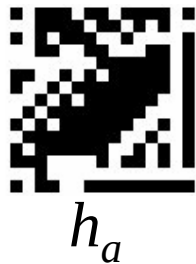
PiSA in Development

- **Platform**
 - Nokia N70 Smartphone
 - Symbian OS 8.1
 - Programming in Unix/Linux
 - Can use Java, C, C++ development kit
- **Technical Approaches**
 - Built with extension of **SiB** techniques
 - McCune, Perrig, Reiter @ IEEE S&P 2005
 - Feasible on modern mobile phones
 - Random Art
 - A visual hash function that produces a random and unique “conceptual” image (art) for any input (string)
 - The image (art) could be easily recognizable by human vision

Authenticating a Public Key with SiB

Alice

$$h_a \leftarrow \text{SHA1}(PK_A)$$



h_a

$\xrightarrow{h_a}$
(visual by camera)

$\xrightarrow{PK_A}$
(Wireless)

Bob

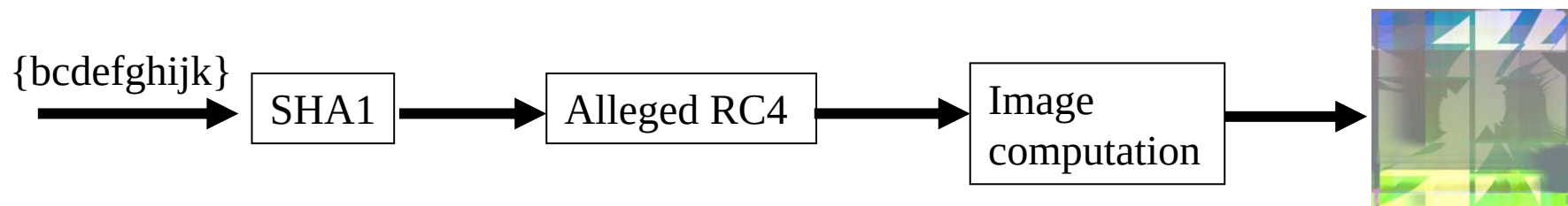


$$h' \leftarrow \text{SHA1}(PK_A)$$

if ($h' \neq h_a$) : abort

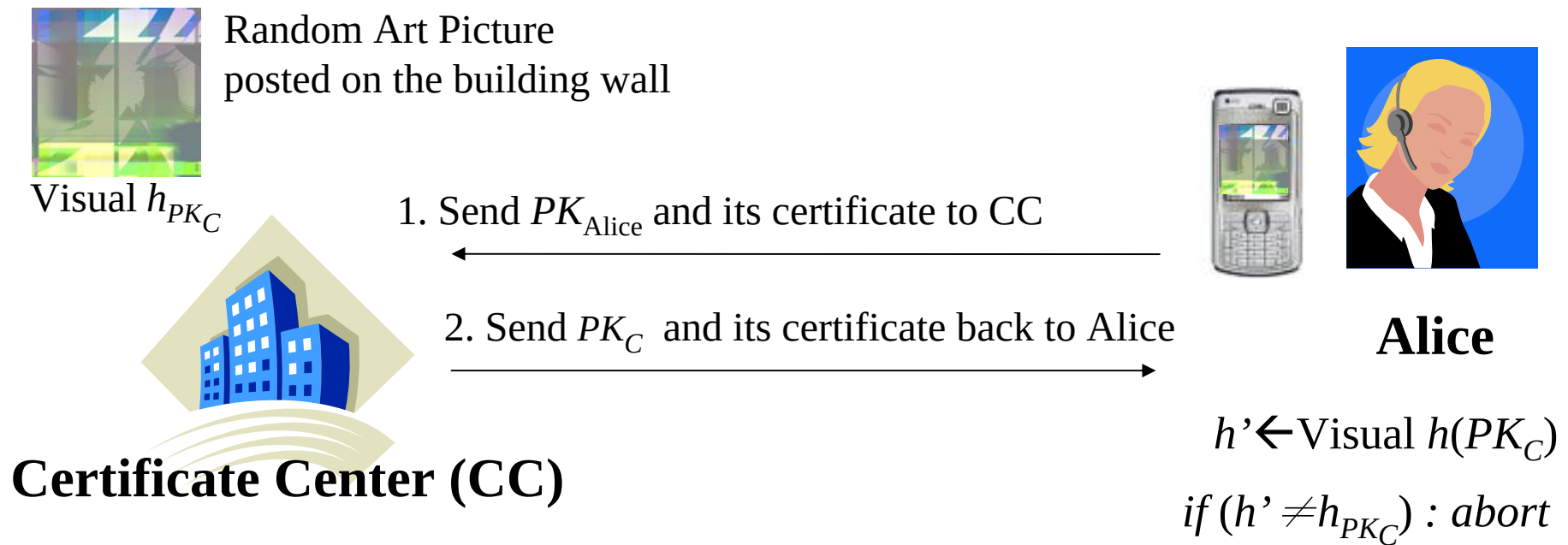
Visual Hash using Random Art

- Input: a seed for pseudo-random number generator
- Computation: generate a random mathematical expression and evaluation for each pixel
- Output: a random art image



Demo 1

Verify the building by using RandomArt



Demo 2

Authenticate other users in the same building with RandomArt



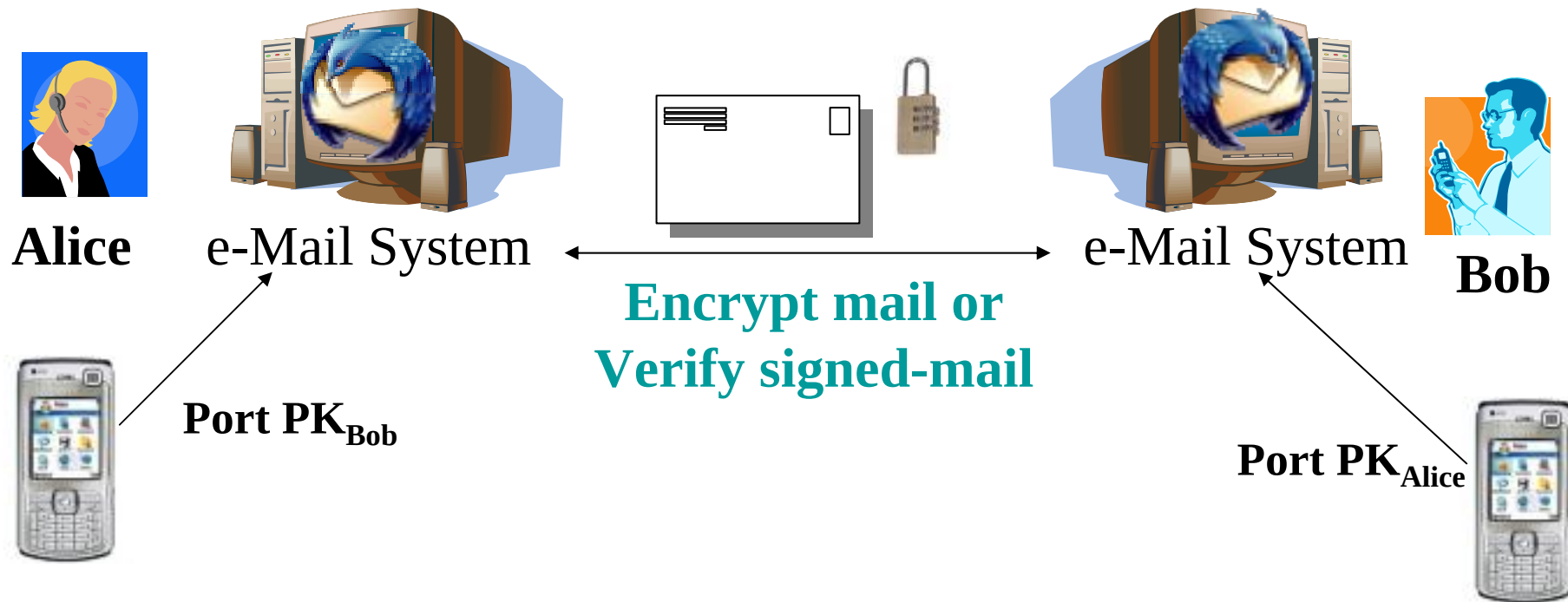
Demo 3

Users do key exchange by using SiB



Potential Application

PGP-like mail system



Conclusions

- TWISC is to serve as an R&D resource center to enhance technical competence, including Ubicomp security
- TWISC is to enlarge human resource capacity and promote public awareness in information security
- TWISC is to be a window for university-industry partnership and international collaboration in information security
- TWISC represents integrated research capabilities, aiming to establish itself as a **power house** in information security, working closely with other renowned centers or laboratories worldwide

